

AYMANE AIT DADS

ML Research Intern | LLM Fine-Tuning · Statistical Learning · Deep Neural Networks

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PROFESSIONAL SUMMARY

Achieved top 10% public leaderboard (0.80/1.0) in the NVIDIA Nemotron Reasoning Kaggle competition (\$106K+ prize pool) by fine-tuning a 30B Mamba-Transformer MoE with LoRA ($r=32$, ~888M trainable parameters) and conducting 20+ single-variable ablation studies to isolate optimal training configurations. Ranked 1st in the EURECOM class leaderboard (0.791 AUC) for AI-generated image detection using CLIP ViT-L/14, with results submitted to the NTIRE 2026 @ CVPR CodaBench international benchmark. Brings hands-on familiarity with PyTorch, Hugging Face Transformers, and the full ML life cycle - spanning model architecture, sequence processing, multi-modal representation, and statistical learning. Available for internship from Summer 2025; holds French student status and is open to USA/Canada placement pending visa sponsorship.

CORE COMPETENCIES

Large Language Models (LLMs) | Statistical Learning | Deep Neural Networks | Sequence Processing | Multi-Modal Learning | Reinforcement Learning | Model Architecture | Natural Language Processing

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Engineered a **Random Forest + K-Means machine learning pipeline** on 100K+ fiber-optic network records - covering upload speed, latency, and jitter - to classify network quality across thousands of nodes.
- Developed a clustering pipeline mapping customer performance profiles to **5 commercial service tiers**, with output adopted directly by the network optimization team for production decisions.
- Reduced manual analysis time by **~60%** through automated feature engineering and systematic model evaluation, accelerating the team's statistical learning workflow.
- Delivered stakeholder presentations translating **model outputs into actionable maintenance recommendations**, demonstrating written and verbal communication skills to non-technical audiences.

PROJECTS

NVIDIA Nemotron Reasoning Challenge

Kaggle Competition · In Progress June 2026 · Top 10% Leaderboard · \$106K+ Prize Pool

- Achieved **top 10% public leaderboard score of 0.80/1.0** (baseline 0.49) by fine-tuning a 30B Mamba-Transformer MoE large language model with LoRA ($r=32$, ~888M trainable parameters) on 7,828 verified chain-of-thought reasoning examples across 6 logic puzzle types.
- Resolved **5 Blackwell GPU (sm_120) incompatibilities** in strict dependency order - including monkey-patched CUDA allocators, stubbed Mamba/Cutlass imports, and a pure-PyTorch RMSNorm fallback - and conducted 20+ single-variable ablation studies confirming completion-only loss and sequence packing impact on out-of-distribution generalization.

PyTorch · Hugging Face Transformers · Unsloth · TRL · PEFT · vLLM · Triton

AI-Generated Image Detection

EURECOM ImSecu + NTIRE 2026 @ CVPR CodaBench · 2025 · Ranked 1st in Class

- Ranked **1st on the EURECOM class private Kaggle leaderboard (0.791 AUC)** by fine-tuning CLIP ViT-L/14 (428M params) with a custom MLP head, dual learning rates, and selective unfreezing of up to 8 transformer blocks; results also submitted to the NTIRE 2026 @ CVPR CodaBench international benchmark.
- Validated that **1-epoch training (0.793 AUC) outperforms 4-epoch (0.767 AUC)** through controlled ablation, then applied a 3-model ensemble with 10-view test-time augmentation on 250K images spanning 25+ generator types for robust multi-modal representation learning.

PyTorch · OpenCLIP · Hugging Face Transformers · FP16 Mixed Precision · Kaggle T4 GPU

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Statistics, Computer Vision, NLP, Cloud Computing, Image Security, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

LLM Fine-Tuning & NLP: LoRA / QLoRA | SFTTrainer | Hugging Face Transformers | Mamba-Transformer MoE | vLLM | Chain-of-Thought | Sentiment Analysis

ML / Deep Learning Frameworks: PyTorch | TensorFlow | Scikit-learn | OpenCLIP | PEFT | TRL | Unsloth

MLOps, Data & Infra: Python | SQL | Bash | Git | Linux | FP16/BF16 Mixed Precision | CUDA Debugging | Pandas | Power BI